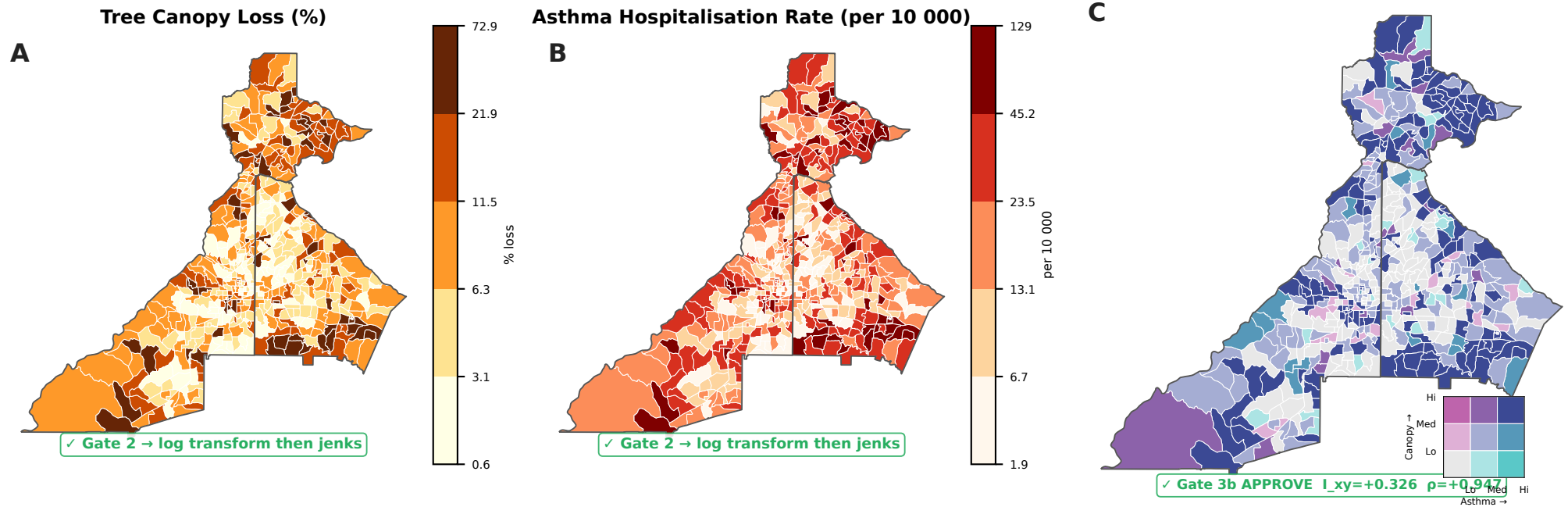




Gate 2 – Classification Diagnostic Engine

Variable	Diagnosis	Prescribed Method	Status
Tree Canopy Loss	heavy right skew	log transform then jenkins	PRESCRIBED
Asthma Hosp. Rate	heavy right skew	log transform then jenkins	PRESCRIBED

Gate 3b — Bivariate Spatial Cross-Correlation

Statistic	Value	p-value	Threshold	Decision
Bivariate Moran's I_{xy}	+0.3262	0.0050	$ I_{xy} > 0.15$	✓ PASS
Spearman's ρ	+0.9471	0.00e+00	$ \rho > 0.20$	✓ PASS

LLM proposed: *jenks (naïve)*. Gate 2 overrides both $\rightarrow$ log-transform first, then Jenks on residuals.

199 permutations under H_0 : no spatial cross-association. Pseudo p -value = $(M+1)/(R+1)$. Bivariate choropleth unlocked.